fuzzy number is a convex fuzzy subset, which has a normalized membership function and represents an interval of confidence.

We remark that the fuzzy subset, with subset $[0,\infty)$ and the associate fuzzy membership function $\mu(x) = 1 - e^{-x}$, is considered to be normal. Although this membership function does not attain the maximum value 1, it approaches 1 as a limit. We consider all such fuzzy subsets as normal in this book.

For the arithmetic of fuzzy numbers, there is a general rule.

The General Rule. Let S_x and S_y be two fuzzy subsets of the universe set **S**, $Z \subseteq R$, and consider a two-variable extended function

$$F: S_x \times S_x \to Z.$$

Let S_z be the image of F, which is a fuzzy subset of Z as discussed above, and $\mu_{S_x}(x)$, $\mu_{S_y}(y)$, and $\mu_{S_z}(z)$ be the associate membership functions. Given $\mu_{S_x}(x)$ and $\mu_{S_y}(y)$, we define

$$\mu_{S_z}(z) = \sup_{z=F(x,y)} \{ \mu_{S_x}(x) \wedge \mu_{S_y}(y) \}.$$

The reason for choosing the smaller value in this definition is that when one has two different degrees of confidence about two events, then the confidence about both events together is lower. Using the α-cut notation, this is equivalent to the following:

$$\begin{aligned}(S_z)_{\overline{\alpha}} &= F(\ (S_x)_{\overline{\alpha}},\ (S_y)_{\overline{\alpha}}\ ) \\ &= \{\ z \in S_z \mid z = F(x,y), x \in\ (S_x)_{\overline{\alpha}}, y \in\ (S_y)_{\overline{\alpha}}\ \}.\end{aligned}$$

Now, given two fuzzy numbers $\tilde{x}$ and $\tilde{y}$, which are two normal and convex fuzzy subsets $S_{\tilde{x}}$ and $S_{\tilde{y}}$ with membership functions $\mu_{S_{\tilde{x}}}$ and $\mu_{S_{\tilde{y}}}$, respectively, consider a two-variable function

$$F: X \times Y \to Z$$

defined by $z = F(x,y)$, $x \in S_{\tilde{x}}$, $y \in S_{\tilde{y}}$, via the general operation rule. We only discuss the fuzzy number arithmetic for addition, subtraction, multiplication, division, minimum, and maximum operations. These results will be needed in the next chapter.

(1) Addition. Let $z = F(x,y) = x + y$. Then $\tilde{z} = \tilde{x} + \tilde{y}$ with

$$S_{\tilde{z}} = \{\ z \in Z \mid z = x + y, x \in\ S_{\tilde{x}}, y \in\ S_{\tilde{y}}\ \}$$

and

$$\mu_{S_{\tilde{z}}}(z) = \sup_{z=x+y} \{\ \mu_{S_{\tilde{x}}}(x) \wedge \mu_{S_{\tilde{y}}}(y)\ \}.$$

In the α-cut notation:

$$(S_{\tilde{z}})_{\overline{\alpha}} = F(\ (S_{\tilde{x}})_{\overline{\alpha}},\ (S_{\tilde{y}})_{\overline{\alpha}}\ ) = (S_{\tilde{x}})_{\overline{\alpha}} + (S_{\tilde{y}})_{\overline{\alpha}}.$$

Example 1.12. Let $\tilde{x}$ and $\tilde{y}$ be such that

$$S_{\tilde{x}} = [-5,1], \qquad S_{\tilde{y}} = [-5,12],$$

with associate membership functions shown in Figure 1.16:

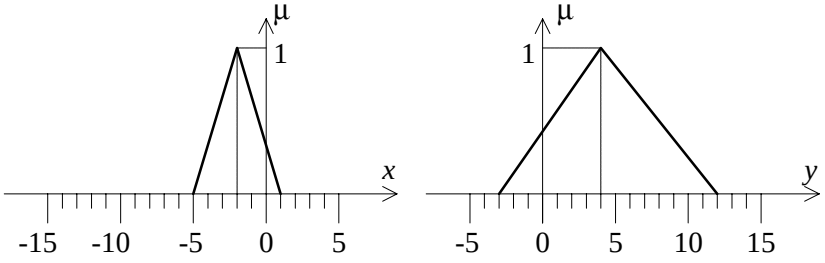


Figure 1.16 Two membership functions.

$$\mu_{S_{\tilde{x}}}(x) = \begin{cases} \dfrac{x}{3} + \dfrac{5}{3}, & -5 \le x \le -2, \\ -\dfrac{x}{3} + \dfrac{1}{3}, & -2 \le x \le 1, \end{cases}$$

and

$$\mu_{S_{\tilde{y}}}(y) = \begin{cases} 0 & -5 \le y \le -3, \\ \dfrac{y}{7} + \dfrac{3}{7} & -3 \le y \le 4, \\ -\dfrac{y}{8} + \dfrac{12}{8} & 4 \le y \le 12. \end{cases}$$

Then, by the general operation rule we have

$$S_{\tilde{z}} = S_{\tilde{x}} + S_{\tilde{y}} = [-5,1] + [-5,12] = [-10,13]$$

and, by comparing $\mu_{S_{\tilde{x}}}(x)$ and $\mu_{S_{\tilde{y}}}(y)$ pointwise, we obtain

$$\mu_{S_{\tilde{z}}}(z) = \sup_{z=x+y} \{ \mu_{S_{\tilde{x}}}(x) \wedge \mu_{S_{\tilde{y}}}(y) \}$$

$$= \begin{cases} 0, & -10 \le z \le -8, \\ \dfrac{z}{10} + \dfrac{8}{10}, & -8 \le z \le 2, \\ -\dfrac{z}{11} + \dfrac{13}{11}, & 2 \le z \le 13. \end{cases}$$

Here, it is clear that the general "sup" rule does not yield the explicit formulas easily. In contrast, this resulting explicit formula of $\mu_{S_{\tilde{z}}}(z)$ can be easily obtained by using the equivalent α-cut operation as follows.

In the α-cut notation, for any α value, the α-cut of $S_{\tilde{x}}$ is obtained by letting $\alpha = x/3 + 5/3$ and $\alpha = -x/3 + 1/3$, respectively, which give $x_1 = 3\alpha - 5$ and $x_2 = -3\alpha + 1$, as shown in (the enlarged) Figure 1.17. Hence, the projection interval is

$$(S_{\tilde{x}})_{\overline{\alpha}} = [x_1, x_2] = [3\alpha - 5, -3\alpha + 1].$$

Similarly,

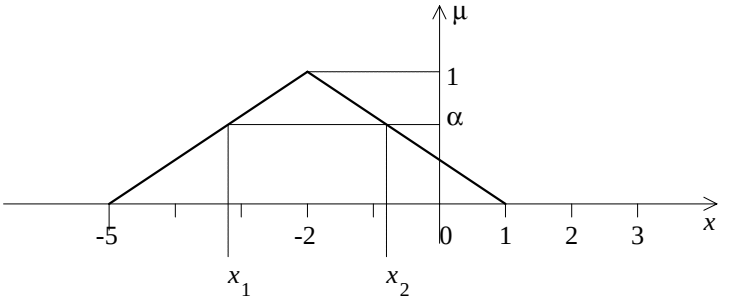


Figure 1.17 The α-cut of the membership function $\mu_{S_{\tilde{x}}}(x)$.

$$(S_{\tilde{y}})_{\overline{\alpha}} = [7\alpha - 3, -8\alpha + 12],$$

so that

$$(S_{\tilde{z}})_{\overline{\alpha}} = (S_{\tilde{x}})_{\overline{\alpha}} + (S_{\tilde{y}})_{\overline{\alpha}} = [10\alpha - 8, -11\alpha + 13].$$

Setting $z_1 = 10\alpha - 8$ and $z_2 = -11\alpha + 13$ gives $\alpha = z_1/10 + 8/10$ and $\alpha = -z_2/11 + 12/11$, which yield the membership function shown in Figure 1.18.

Taking $S_{\tilde{z}} = [-10,13]$ into account, we finally arrive at

$$\mu_{S_{\tilde{z}}}(z) = \begin{cases} 0, & -10 \le z \le -8, \\ \dfrac{z}{10} + \dfrac{8}{10}, & -8 \le z \le 2, \\ -\dfrac{z}{11} + \dfrac{13}{11}, & 2 \le z \le 13. \end{cases}$$

(2) Subtraction. Let $z = F(x,y) = x - y$. Then $\tilde{z} = \tilde{x} - \tilde{y}$ with

$$S_{\tilde{z}} = \{ z \in Z \mid z = x - y, x \in S_{\tilde{x}}, y \in S_{\tilde{y}} \}$$

and

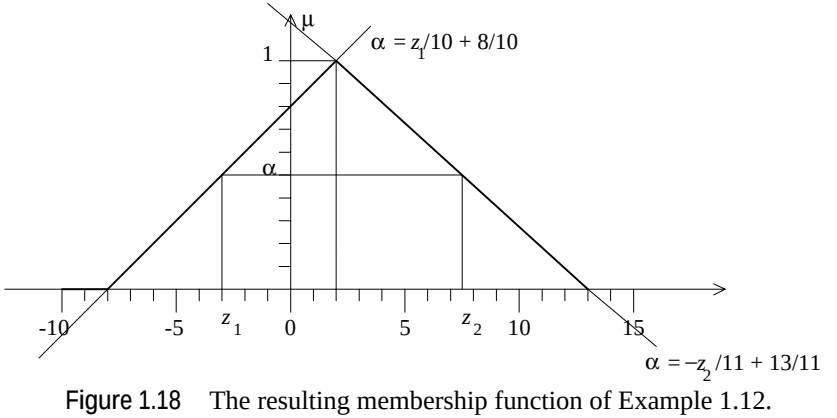


Figure 1.18 The resulting membership function of Example 1.12.

$$\mu_{S_{\tilde{z}}}(z) = \sup_{z=x-y} \{ \mu_{S_{\tilde{x}}}(x) \wedge \mu_{S_{\tilde{y}}}(y) \}.$$

In the α-cut notation:

$$(S_{\tilde{z}})_{\overline{\alpha}} = F((S_{\tilde{x}})_{\overline{\alpha}}, (S_{\tilde{y}})_{\overline{\alpha}}) = (S_{\tilde{x}})_{\overline{\alpha}} - (S_{\tilde{y}})_{\overline{\alpha}}.$$

Example 1.13. Let $\tilde{x}$ and $\tilde{y}$ be such that

$$S_{\tilde{x}} = [0,20], \qquad S_{\tilde{y}} = [0,10],$$

with the membership functions

$$\mu_{S_{\tilde{x}}}(x) = \begin{cases} 0, & 0 \le x \le 7, \\ \dfrac{x}{7} - 1, & 7 \le x \le 14, \\ -\dfrac{x}{5} + \dfrac{19}{5}, & 14 \le x \le 19, \\ 0, & 19 \le x \le 20, \end{cases}$$

and

$$\mu_{S_{\tilde{y}}}(y) = \begin{cases} 0, & 0 \le y \le 3, \\ \dfrac{y}{2} - \dfrac{3}{2}, & 3 \le y \le 5, \\ -\dfrac{y}{5} + 2, & 5 \le y \le 10. \end{cases}$$

Then we have, via the interval arithmetic,

$$S_{\tilde{z}} = S_{\tilde{x}} - S_{\tilde{y}} = [-10,20],$$

with

$$\mu_{S_{\tilde{z}}}(z) = \begin{cases} 0, & -10 \le z \le -3, \\ \dfrac{z}{12} + \dfrac{3}{12} & -3 \le z \le 9, \\ -\dfrac{z}{7} + \dfrac{16}{7} & 9 \le z \le 16, \\ 0, & 16 \le z \le 20. \end{cases}$$

(3) Multiplication. Let $z = F(x,y) = x \cdot y$. Then $\tilde{z} = \tilde{x} \cdot \tilde{y}$ with

$$S_{\tilde{z}} = \{ z \in Z \mid z = x\,y, x \in S_{\tilde{x}}, y \in S_{\tilde{y}} \}$$

and

$$\mu_{S_{\tilde{z}}}(z) = \sup_{z=x \cdot y} \{ \mu_{S_{\tilde{x}}}(x) \wedge \mu_{S_{\tilde{y}}}(y) \}.$$

In the α-cut notation:

$$(S_{\tilde{z}})_{\overline{\alpha}} = F((S_{\tilde{x}})_{\overline{\alpha}}, (S_{\tilde{y}})_{\overline{\alpha}}) = (S_{\tilde{x}})_{\overline{\alpha}} \cdot (S_{\tilde{y}})_{\overline{\alpha}}.$$

Example 1.14. Let $\tilde{x}$ and $\tilde{y}$ be such that

$$S_{\tilde{x}} = [2,5], \qquad S_{\tilde{y}} = [3,6],$$

with the membership functions

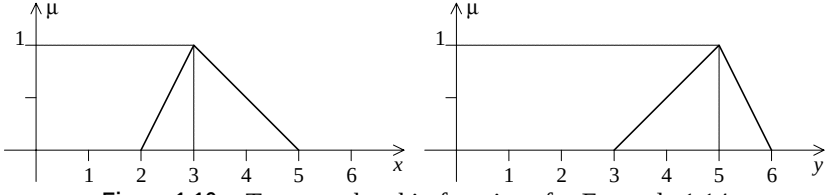


Figure 1.19 Two membership functions for Example 1.14.

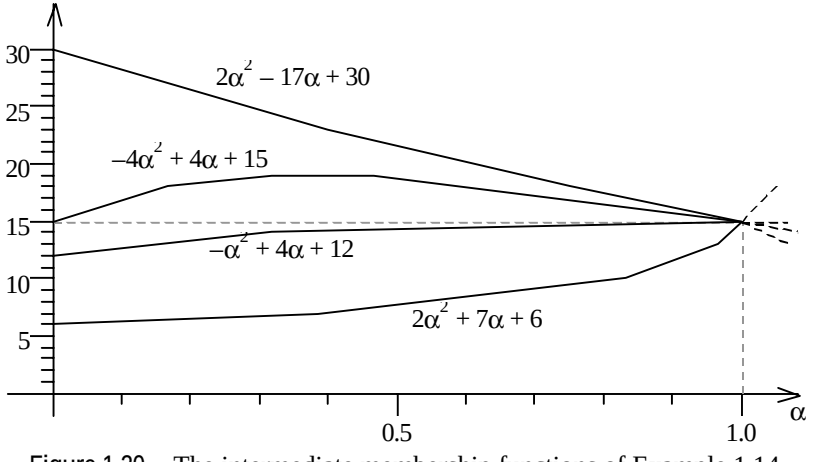


Figure 1.20 The intermediate membership functions of Example 1.14.

$$\mu_{S_{\tilde{x}}}(x) = \begin{cases} x-2, & 2 \le x \le 3, \\ -\dfrac{x}{2}+\dfrac{5}{2}, & 3 \le x \le 5, \end{cases}$$

and

$$\mu_{S_{\tilde{y}}}(y) = \begin{cases} \dfrac{y}{2}-\dfrac{3}{2}, & 3 \le y \le 5, \\ -y+6, & 5 \le y \le 6, \end{cases}$$

as shown in Figure 1.19.

In the α-cut notation, for any α value, letting

$$\alpha = x - 2 \qquad \text{and} \qquad \alpha = -\frac{x}{2} + \frac{5}{2}$$

gives

$$x_1 = \alpha + 2 \qquad \text{and} \qquad x_2 = -2\alpha + 5,$$

so that

$$(S_{\tilde{x}})_{\overline{\alpha}} = [\, \alpha + 2, -2\alpha + 5 \,].$$

Similarly,

$$(S_{\tilde{y}})_{\overline{\alpha}} = [\, 2\alpha + 3, -\alpha + 6 \,].$$

It then follows that

$$S_{\tilde{z}} = [6,30]$$

and

$$\begin{aligned}(S_{\tilde{z}})_{\overline{\alpha}} &= (S_{\tilde{x}})_{\overline{\alpha}} \cdot (S_{\tilde{y}})_{\overline{\alpha}} \\ &= [\alpha + 2, -2\alpha + 5\,] \cdot [\, 2\alpha + 3, -\alpha + 6\,] \\ &= [\underline{p}(\alpha), \overline{p}(\alpha)\,],\end{aligned}$$

where

$$\begin{aligned}\underline{p}(\alpha) &= min\{2\alpha^2+7\alpha+6, -\alpha^2+4\alpha+12, -4\alpha^2+4\alpha+15, 2\alpha^2-17\alpha+30\}, \\ \overline{p}(\alpha) &= max\{2\alpha^2+7\alpha+6, -\alpha^2+4\alpha+12, -4\alpha^2+4\alpha+15, 2\alpha^2-17\alpha+30\},\end{aligned}$$

with the curves shown in Figure 1.20.

Hence, $\underline{p}(\alpha) = 2\alpha^2+7\alpha+6$, and $\overline{p}(\alpha) = 2\alpha^2-17\alpha+30$, so that

$$\begin{aligned}(S_{\tilde{z}})_{\overline{\alpha}} &= [\underline{p}(\alpha), \overline{p}(\alpha)\,] \\ &= [\, 2\alpha^2+7\alpha+6,\ 2\alpha^2-17\alpha+30\,].\end{aligned}$$

Let, moreover,

$$z_1 = 2\alpha^2+7\alpha+6,$$
$$z_2 = 2\alpha^2-17\alpha+30.$$

We solve them for α, subject to $0 \le \alpha \le 1$, and obtain

$$\alpha = \frac{-7+\sqrt{1+8z_1}}{4},$$

or

$$\alpha = \frac{17-\sqrt{49+8z_2}}{4}.$$

Consequently, we have

$$\mu_{S_{\tilde{z}}}(z) = \begin{cases} \dfrac{-7+\sqrt{1+8z}}{4}, & 6 \le z \le 15, \\ \dfrac{17-\sqrt{49+8z}}{4}, & 15 \le z \le 30, \end{cases}$$

as shown in Figure 1.21.

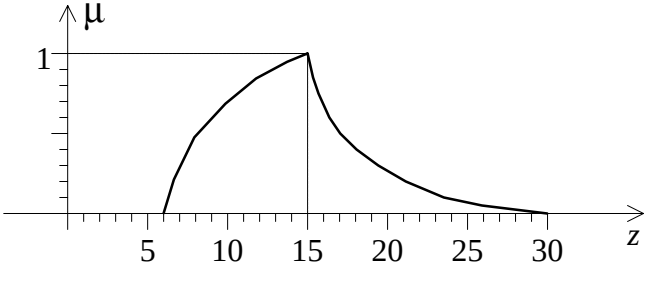


Figure 1.21 The resulting membership function of Example 1.14.

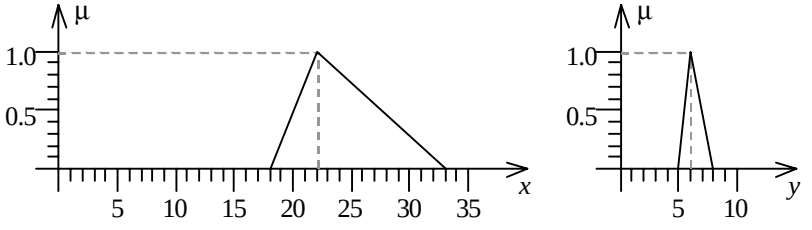


Figure 1.22 The two membership functions for Example 1.15.

(4) Division. Let $z = F(x,y) = \dfrac{x}{y}$. Then $\tilde{z} = \dfrac{\tilde{x}}{\tilde{y}}$ with

$$S_{\tilde{z}} = \{\, z \in Z \mid z = x/y,\ x \in S_{\tilde{x}},\ y \in S_{\tilde{y}} \,\}$$

and

$$\mu_{S_{\tilde{z}}}(z) = \sup_{z=x/y} \{\, \mu_{S_{\tilde{x}}}(x) \wedge \mu_{S_{\tilde{y}}}(y) \,\}.$$

In the α-cut notation:

$$(S_{\tilde{z}})_{\overline{\alpha}} = F(\, (S_{\tilde{x}})_{\overline{\alpha}},\ (S_{\tilde{y}})_{\overline{\alpha}} \,) = \frac{(S_{\tilde{x}})_{\overline{\alpha}}}{(S_{\tilde{y}})_{\overline{\alpha}}}.$$

Example 1.15. Let $\tilde{x}$ and $\tilde{y}$ be such that

$$S_{\tilde{x}} = [18,33], \qquad S_{\tilde{y}} = [5,8],$$

with the membership functions

$$\mu_{S_{\tilde{x}}}(x) = \begin{cases} \dfrac{x}{4} - \dfrac{18}{4}, & 18 \le x \le 22, \\ -\dfrac{x}{11} + 3, & 22 \le x \le 33, \end{cases}$$

and

$$\mu_{S_{\tilde{y}}}(y) = \begin{cases} y - 5, & 5 \le y \le 6, \\ -\dfrac{y}{2} + 4, & 6 \le y \le 8, \end{cases}$$

as shown in Figure 1.22.

In the α-cut notation, let

$$\alpha = \frac{x}{4} - \frac{18}{4} \qquad \text{and} \qquad \alpha = -\frac{x}{11} + 3.$$

We obtain

$$x_1 = 4\alpha + 18 \qquad \text{and} \qquad x_2 = -11\alpha + 33,$$

so that

$$(S_{\tilde{x}})_{\overline{\alpha}} = [\, 4\alpha + 18,\ -11\alpha + 33 \,].$$

Similarly,

$$(S_{\tilde{y}})_{\overline{\alpha}} = [\, \alpha + 5,\ -2\alpha + 8 \,].$$

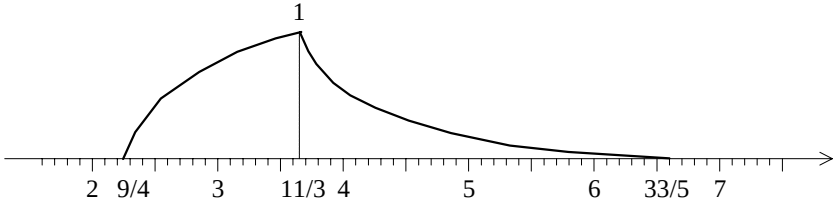


Figure 1.23 The resulting membership function of Example 1.15.

Hence,

$$(S_{\tilde{z}})_{\overline{\alpha}} = \frac{(S_{\tilde{x}})_{\overline{\alpha}}}{(S_{\tilde{y}})_{\overline{\alpha}}} = \frac{[4\alpha+18,-11\alpha+33]}{[\alpha+5,-2\alpha+8]}$$

$$= \left[\frac{4\alpha+18}{-2\alpha+8}, \frac{-11\alpha+33}{\alpha+5}\right].$$

Next, letting

$$z_1 = \frac{4\alpha+18}{-2\alpha+8} \quad \text{and} \quad z_2 = \frac{-11\alpha+33}{\alpha+5}$$

gives

$$\alpha = \frac{8z_1-18}{2z_1+4} \quad \text{and} \quad \alpha = \frac{-5z_2+33}{z_2+11},$$

so that

$$\mu_{S_{\tilde{z}}}(z) = \begin{cases} \dfrac{8z-18}{2z+4}, & \dfrac{9}{4} \le z \le \dfrac{11}{3}, \\ \dfrac{-5z+33}{z+11}, & \dfrac{11}{3} \le z \le \dfrac{33}{5}, \end{cases}$$

as shown in Figure 1.23.

(5) Minimum and Maximum. Let $\tilde{x}$ and $\tilde{y}$ be two fuzzy numbers with $S_{\tilde{x}} = [a,b]$ and $S_{\tilde{y}} = [c,d]$, and with membership functions $\mu_{S_{\tilde{x}}}(x)$ and $\mu_{S_{\tilde{y}}}(y)$, respectively. For any α value, let us denote

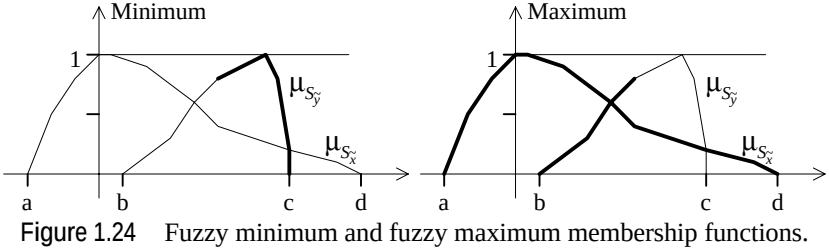


Figure 1.24 Fuzzy minimum and fuzzy maximum membership functions.

$$(S_{\tilde{x}})_{\overline{\alpha}} = [\underline{x}(\alpha), \overline{x}(\alpha)] \qquad \text{and} \qquad (S_{\tilde{y}})_{\overline{\alpha}} = [\underline{y}(\alpha), \overline{y}(\alpha)].$$

Then, the *fuzzy minimum* of $\tilde{x}$ and $\tilde{y}$ is defined to be $\tilde{z} = \tilde{x} \wedge \tilde{y}$, with

$$S_{\tilde{z}} = [\, min\{a,c\}, min\{b,d\} \,]$$

and

$$\mu_{S_{\tilde{z}}}(z) = \sup_{z=min\{x,y\}} \{ \mu_{S_{\tilde{x}}}(x) \wedge \mu_{S_{\tilde{y}}}(y) \} = \bigvee_{z=x\wedge y} \{ \mu_{S_{\tilde{x}}}(x) \wedge \mu_{S_{\tilde{y}}}(y) \}$$

In the α-cut notation:

$$(S_{\tilde{z}})_{\overline{\alpha}} = (S_{\tilde{x}})_{\overline{\alpha}} \wedge (S_{\tilde{y}})_{\overline{\alpha}} = [\, \underline{x}(\alpha) \wedge \underline{y}(\alpha),\ \overline{x}(\alpha) \wedge \overline{y}(\alpha) \,].$$

Similarly, the *fuzzy maximum* of $\tilde{x}$ and $\tilde{y}$ is defined as $\tilde{z} = \tilde{x} \vee \tilde{y}$, with

$$S_{\tilde{z}} = [\, max\{a,c\}, max\{b,d\} \,]$$

and

$$\mu_{S_{\tilde{z}}}(z) = \bigvee_{z=x\vee y} \{ \mu_{S_{\tilde{x}}}(x) \wedge \mu_{S_{\tilde{y}}}(y) \}.$$

In the α-cut notation:

$$(S_{\tilde{z}})_{\overline{\alpha}} = (S_{\tilde{x}})_{\overline{\alpha}} \vee (S_{\tilde{y}})_{\overline{\alpha}} = [\, \underline{x}(\alpha) \vee \underline{y}(\alpha),\ \overline{x}(\alpha) \vee \overline{y}(\alpha) \,].$$

These two extrema are visualized in Figure 1.24. Graphically, if we draw horizontal lines to cut the two membership functions, then the resulting membership functions have all left-hand or right-hand crossing points, respectively.

Example 1.16. Let $\tilde{x}$ and $\tilde{y}$ be such that

$$S_{\tilde{x}} = [-2,6], \qquad S_{\tilde{y}} = [-4,5],$$

with the membership functions

$$\mu_{S_{\tilde{x}}}(x) = \begin{cases} \dfrac{x}{2}+1, & -2 \le x \le 0, \\ -\dfrac{x}{6}+1, & 0 \le x \le 6, \end{cases}$$

$$\mu_{S_{\tilde{y}}}(y) = \begin{cases} \dfrac{y}{7}+\dfrac{4}{7}, & -4 \le y \le 3, \\ -\dfrac{y}{2}+\dfrac{5}{2}, & 3 \le y \le 5, \end{cases}$$

as shown in Figure 1.25.

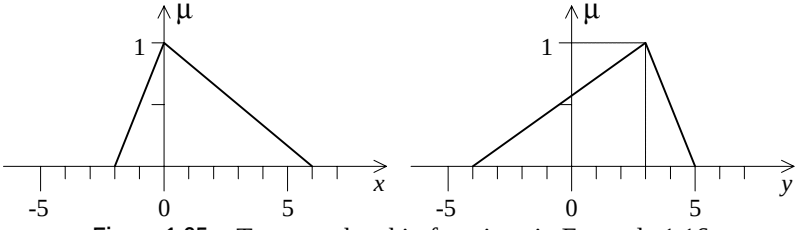


Figure 1.25 Two membership functions in Example 1.16.

In the α-cut notation, following the procedure described above, we obtain

$$(S_{\tilde{x}})_{\overline{\alpha}} = [\, 2\alpha - 2, -6\alpha + 6 \,],$$

and

$$(S_{\tilde{y}})_{\overline{\alpha}} = [\, 7\alpha - 4, -2\alpha + 5 \,],$$

so that

$$\begin{aligned}(S_{\tilde{z}})_{\overline{\alpha}} &= (S_{\tilde{x}})_{\overline{\alpha}} \wedge (S_{\tilde{y}})_{\overline{\alpha}} \\ &= [\, (2\alpha - 2) \wedge (7\alpha - 4), (-6\alpha + 6) \wedge (-2\alpha + 5) \,] \\ &= \begin{cases} [7\alpha - 4, -2\alpha + 5], & 0 \le \alpha \le 1/4, \\ [7\alpha - 4, -6\alpha + 6], & 1/4 \le \alpha \le 2/5, \\ [2\alpha - 2, -6\alpha + 6], & 2/5 \le \alpha \le 1, \end{cases}\end{aligned}$$

in which by matching $2\alpha - 2 = 7\alpha - 4$ and $-6\alpha + 6 = -2\alpha + 5$ we can find the switching points $\alpha = 1/4$ and $\alpha = 2/5$, respectively. It then yields $S_{\tilde{z}} = [-4,5]$, with

$$\mu_{S_{\tilde{z}}}(z) = \begin{cases} -\dfrac{z}{7} + \dfrac{4}{7}, & -4 \le z \le -\dfrac{6}{5}, \\ \dfrac{z}{2} + 1, & -\dfrac{6}{5} \le z \le 0, \\ -\dfrac{z}{6} + 1, & 0 \le z \le \dfrac{9}{2}, \\ -\dfrac{z}{2} + \dfrac{5}{2}, & \dfrac{9}{2} \le z \le 5, \end{cases}$$

as shown in Figure 1.26. Graphically, if we draw horizontal lines to cut the two membership functions, then the resulting membership function has all left-hand crossing points.

Similarly, it can be verified that

$$\begin{aligned}(S_{\tilde{z}})_{\overline{\alpha}} &= (S_{\tilde{x}})_{\overline{\alpha}} \vee (S_{\tilde{y}})_{\overline{\alpha}} \\ &= [\, (2\alpha - 2) \vee (7\alpha - 4), (-6\alpha + 6) \vee (-2\alpha + 5) \,] \\ &= \begin{cases} [2\alpha - 2, -6\alpha + 6], & 0 \le \alpha \le 1/4, \\ [2\alpha - 2, -2\alpha + 5], & 1/4 \le \alpha \le 2/5, \\ [7\alpha - 4, -2\alpha + 5], & 2/5 \le \alpha \le 1, \end{cases}\end{aligned}$$

and

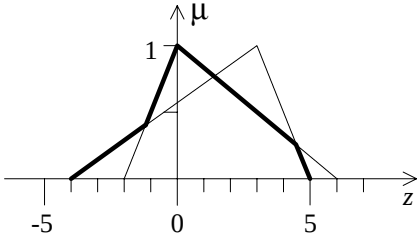


Figure 1.26 The resulting fuzzy minimum membership function.

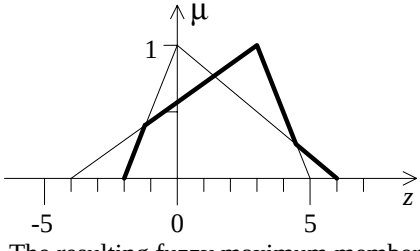


Figure 1.27 The resulting fuzzy maximum membership function.

$$\mu_{S_z}(z) = \begin{cases} \dfrac{z}{2}+1, & -2 \le z \le -\dfrac{6}{5}, \\ \dfrac{z}{7}+\dfrac{4}{7}, & -\dfrac{6}{5} \le z \le 3, \\ -\dfrac{z}{2}+\dfrac{5}{2}, & 3 \le z \le \dfrac{9}{2}, \\ -\dfrac{z}{6}+1, & \dfrac{9}{2} \le z \le 6, \end{cases}$$

as shown in Figure 1.27. Graphically, if we draw horizontal lines to cut the two membership functions, then the resulting membership function has all right-hand crossing points.